CYBERSECURITY RISK ASSESSMENT OF AN ONLINE LEARNING SYSTEM

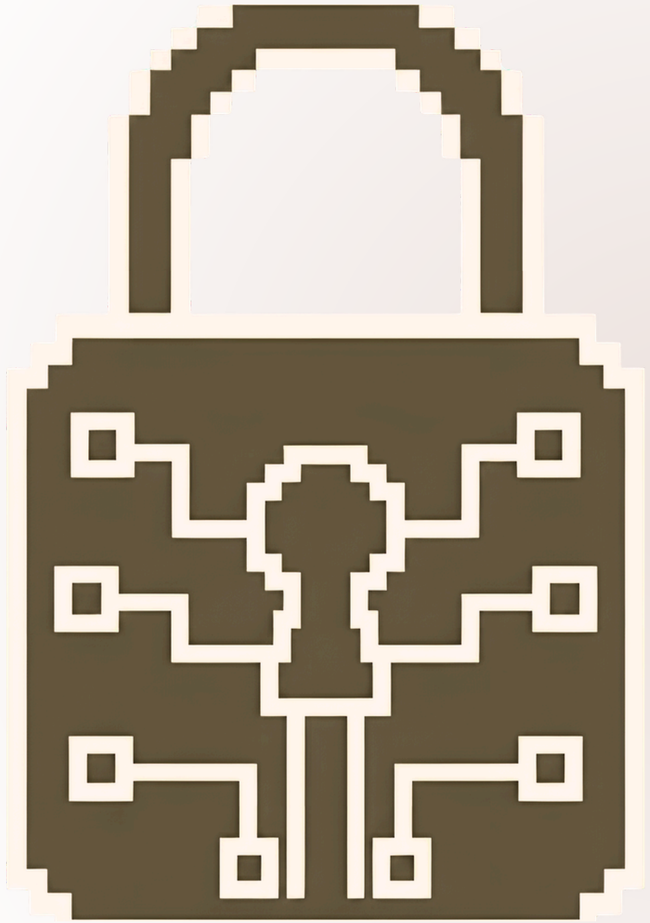
Case Study

Identify Assets

The educational institution relies on an online learning system to manage academic activities, store student information, and provide educational services. This system contains several important assets that must be protected to ensure the confidentiality, integrity, and availability of information.

Student Database

The student database is one of the most valuable assets in the institution because it contains sensitive student information, including personal details, academic grades, and attendance records. Protecting this database is essential to prevent unauthorized access, data theft, or modification of student records. Any security incident affecting the database could compromise students' privacy and reduce trust in the institution.

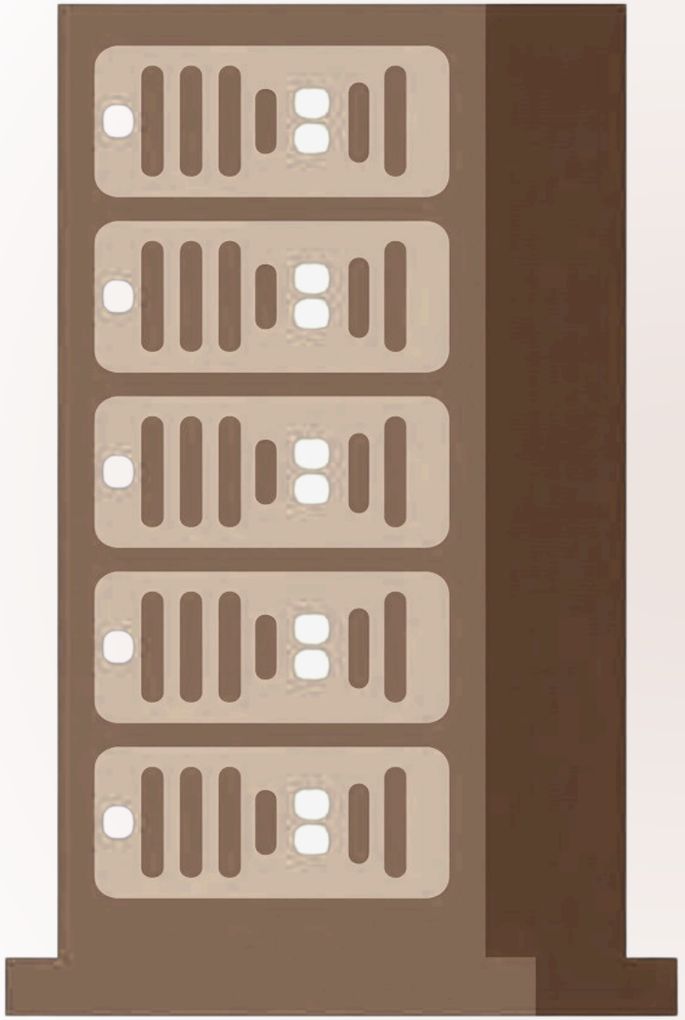


Web Server

The web server is responsible for hosting the online learning system and allowing students and employees to access educational services. It plays an important role in maintaining the availability of the system. If the server becomes compromised or unavailable, students and instructors may be unable to access courses, academic materials, and other online services. Therefore, protecting the web server is necessary to ensure continuous and secure access.

Course Files

Course files are important educational assets because they contain learning materials, lecture documents, assignments, and other academic resources. These files support the learning process and must remain accessible to authorized students and instructors. Unauthorized modification, deletion, or loss of course files could disrupt academic activities and affect the quality of educational services.



Identify Threats and Vulnerabilities

The institution’s online learning system faces several cybersecurity threats due to existing security weaknesses. Identifying these threats and vulnerabilities is essential to understanding the potential risks and protecting sensitive information.



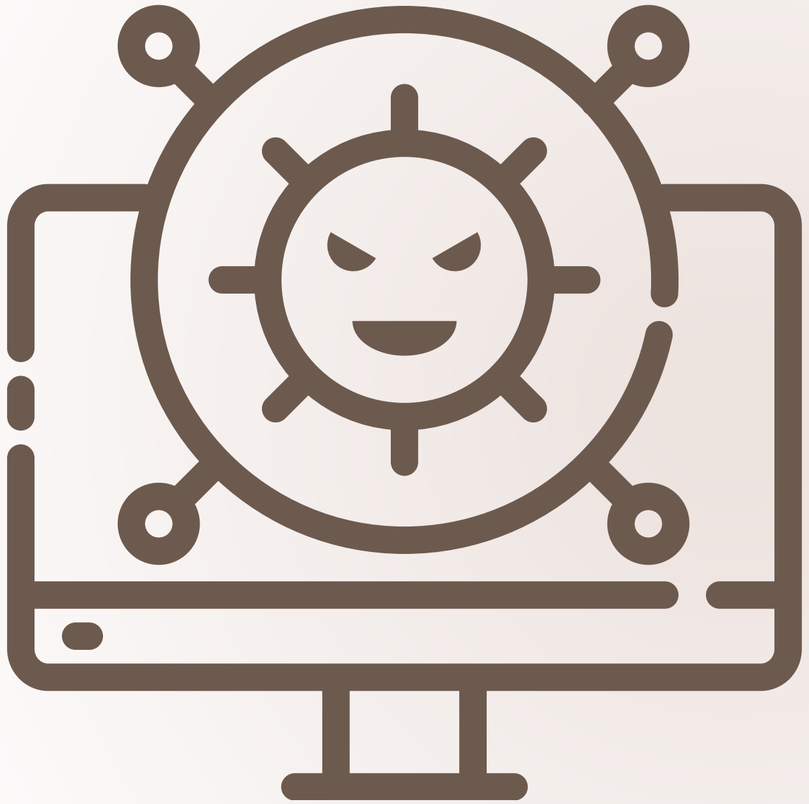
Unauthorized Access Due to Weak or Shared Passwords

One of the main security threats is unauthorized access to the online learning system. Some employees use weak or shared passwords, which increases the possibility of unauthorized individuals gaining access to their accounts.

This vulnerability may allow attackers to access confidential student information, modify academic records, or misuse the system. As a result, the institution may experience privacy violations and loss of trust.

Malware Attacks Due to Missing Security Updates

Another significant threat is malware infection or exploitation of the web server. If the server does not receive regular security updates, it may contain known vulnerabilities that attackers could exploit.
These weaknesses could allow malicious software to affect the server, interrupt online learning services, or compromise stored information. This threat may negatively affect the availability and security of the institution’s system.



Data Theft Due to Insecure Remote Access

The online learning system may also face data theft risks because remote access is enabled. If remote access is not properly secured, unauthorized individuals may attempt to gain access to the institution’s network or server.
This vulnerability could expose sensitive student records, including personal information, grades, and attendance data. Such incidents may lead to privacy breaches and damage the institution’s reputation

Qualitative Risk Assessment

- Qualitative Risk Assessment is the process of evaluating potential security risks based on their likelihood of occurring and the impact they may have on the organization. In this project, three risks related to the online learning system are assessed using a scale from 1 to 10.

1. The risk score is calculated using the following formula:

Risk Score = Likelihood × Impact

The risk levels are classified according to the project guidelines:

- Low Risk: 1–20
- Medium Risk: 21–50
- High Risk: 51–100

2. Risk Assessment Table

Identified Risk Likelihood (1–10) Impact (1–10) Risk Score Risk Level
Unauthorized Access 8 9 72 High
Server Exploitation 7 9 63 High
Loss of Student Records 5 10 50 Medium

1. Unauthorized Access

Unauthorized access may occur because some employees use weak or shared passwords. Attackers may gain access to the online learning system and view or misuse sensitive student information

Risk Score = 8 × 9 = 72 (High)

2. Server Exploitation

The server may be vulnerable to cyberattacks because it is missing important security updates. Attackers could exploit these vulnerabilities to access the server, steal information, or interrupt system services.

Risk Score = 7 × 9 = 63 (High)

3. Loss of Student Records

Student records may be lost due to cyberattacks, system failures, or accidental deletion. This could affect student information, grades, and attendance records.

Risk Score = 5 × 10 = 50 (Medium)

Qualitative Risk Assessment Conclusion

Based on the assessment, unauthorized access is the highest-priority risk, with a score of 72, followed by server exploitation, with a score of 63. Both risks are classified as High and require immediate attention.

Loss of student records has a score of 50, which is classified as Medium. However, it is still important to protect student information and ensure that appropriate security measures are implemented.

Note: Likelihood and impact ratings are reasonable assessment assumptions based on the case study.

4. Quantitative Risk Assessment

Quantitative Risk Assessment estimates the potential financial loss resulting from a security incident. In this project, the calculations are based on the hypothetical values provided in the project requirements.
The selected risk is Unauthorized Access, which may expose sensitive student information stored in the online learning system.

Given Data

Item Value
Affected Asset Value SAR 30,000
Exposure Factor (EF) 20% (0.20)
Annual Rate of Occurrence (ARO) 2 times per year

1. Single Loss Expectancy (SLE)

Single Loss Expectancy (SLE) represents the estimated financial loss resulting from one security incident.

Formula:

SLE = Asset Value × Exposure Factor

Calculation:

SLE = 30,000 × 0.20

SLE = SAR 6,000

This means that the estimated financial loss from one unauthorized access incident is SAR 6,000.

Annualized Loss Expectancy (ALE) .2 ا

.Annualized Loss Expectancy (ALE) represents the estimated financial loss over one year

:Formula

$$ALE = SLE \times ARO$$

:Calculation

$$ALE = 6{,}000 \times 2$$

$$ALE = SAR\ 12{,}000\ per\ year$$

This means that the organization may experience an estimated financial loss of SAR 12,000 annually if the incident occurs .twice per year

Quantitative Risk Assessment Conclusion

.The quantitative assessment shows that a single unauthorized access incident may result in a financial loss of SAR 6,000 .Based on an Annual Rate of Occurrence of two incidents per year, the estimated annual financial loss is SAR 12,000

These results demonstrate the importance of implementing effective security measures, such as multi-factor authentication, strong passwords, regular security updates, and continuous monitoring, to reduce the likelihood of .unauthorized access and minimize potential financial losses

Risk Management Responses

Organizations can use four main strategies to manage cybersecurity risks in an online learning system

1- Risk Mitigation
Implementing security measures to reduce the likelihood and impact of cyber threats. For example, using multi-factor authentication (MFA), data encryption, and regular software updates to protect student and instructor accounts.

2- Risk Avoidance
Eliminating activities that may expose the online learning system to unnecessary risks. For example, disabling outdated services and removing unnecessary system features that could create security vulnerabilities.

3- Risk Transfer
Transferring some cybersecurity risks to a third party through cybersecurity insurance or outsourcing specific security services to specialized providers.

4-Risk Acceptance
Accepting low-level risks when the cost of implementing security measures is higher than the potential impact. However, these risks should be documented and continuously monitored.

These strategies help ensure the confidentiality, integrity, and availability of the online learning system.

Security Controls

To protect the online learning system from cybersecurity threats, the following security controls are recommended

1- Multi-Factor Authentication (MFA)
Require users to verify their identity using more than one authentication method to prevent unauthorized access.

2- Data Encryption
Encrypt sensitive information, including student records, personal information, and login credentials, both during transmission and storage.

3- Access Control
Apply Role-Based Access Control (RBAC) to ensure that students, instructors, and administrators can only access the information and resources required for their roles.

4-Regular Security Updates
Keep the learning platform, operating systems, and applications updated to fix known vulnerabilities and prevent cyberattacks.

5-Firewall and Intrusion Detection Systems
Use firewalls and Intrusion Detection Systems (IDS) to monitor network traffic, detect suspicious activities, and block unauthorized access.

6-Data Backup and Recovery
Perform regular backups and establish a recovery plan to restore educational data and system services in case of cyberattacks or data loss.

7-Security Awareness Training
Provide students and staff with cybersecurity awareness training to help them recognize phishing emails, suspicious links, and other cyber threats.

Implementing these security controls will strengthen the protection of the online learning system and reduce the likelihood of security incidents.

Vulnerability Scanning Plan

Authorized Scope: Focus exclusively on the institution's web server and associated online learning infrastructure to ensure targeted and relevant assessment

Timing: Schedule scans during off-peak hours (e.g., weekends or late nights) to prevent service interruptions for students and staff

Items to Check: Open ports and exposed services, Weak/default accounts, Missing patches, Insecure configurations

- **Documentation & Rules:**

1. Record and report all discovered findings in a structured, formal report
2. **Strict Policy:** Authorization is for scanning and detection only—do not exploit any identified vulnerabilities

Interpret Scan Results

Scan Finding 1: Critical missing security patches identified on the main web server, creating direct pathways for remote code execution

Classification: High

Scan Finding 2: Insecure remote access configurations combined with weak a default employee password practices

Classification: Medium

Priority & Rationale: Fix Finding 1 first. Unpatched software gives attackers direct control over the server, posing the highest risk of complete data theft or service disruption

Residual Risk and Monitoring

- **Control Verification:** Conduct follow-up verification scans to ensure patches were applied successfully and ports are properly secured.
- **Security Monitoring:** Continuously review system event logs to detect and alert on unusual login attempts or suspicious activity.
- **Residual Risk Management:** Acknowledge minor lingering risks (such as zero-days) and maintain ongoing threat readiness and policy updates.

Conclusions

HIGHEST-PRIORITY RISK

- **Unpatched web server vulnerabilities**
- **Weak or shared employee credentials**
- **Unrestricted remote access points**

From a cybersecurity consultancy perspective, the comprehensive risk assessment of the institution's online learning system reveals that the highest-priority exposure stems from a compounding chain of vulnerabilities. The presence of unpatched web server vulnerabilities, combined with weak or shared administrative credentials and open remote access, creates an unstable security posture. These technical flaws collectively expose core operational assets—specifically sensitive student records, academic grades, attendance logs, and proprietary course materials—to severe threats of unauthorized data exfiltration, system compromise, and malicious disruption of academic continuity. Without immediate intervention, these high-priority vectors serve as primary entry points for threat actors aiming to exploit the institution's digital infrastructure.

Conclusions

MOST IMPORTANT NEXT ACTION

To systematically neutralize these critical vulnerabilities and safeguard the institution's digital ecosystem, the primary and most vital immediate action required is the strict enforcement of advanced access controls and robust authentication mechanisms. This involves mandating Multi-Factor Authentication (MFA) across all administrative and user accounts, alongside the implementation of strict, unique password policies. Concurrently, the technical team must execute an emergency deployment of missing server security patches, temporarily restrict or secure all remote access entry points, and establish a continuous, non-exploitative vulnerability-scanning routine. Pairing these technical remediations with ongoing security monitoring and proactive threat intelligence will ensure long-term system resilience, protect data integrity, and maintain a secure, trusted learning environment for all stakeholders.

THANK YOU ALL

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